**EE2001 Electrical Circuits- I:**

Continuation to:

# Circuit Elements

16 October 2023, Monday

# Outline

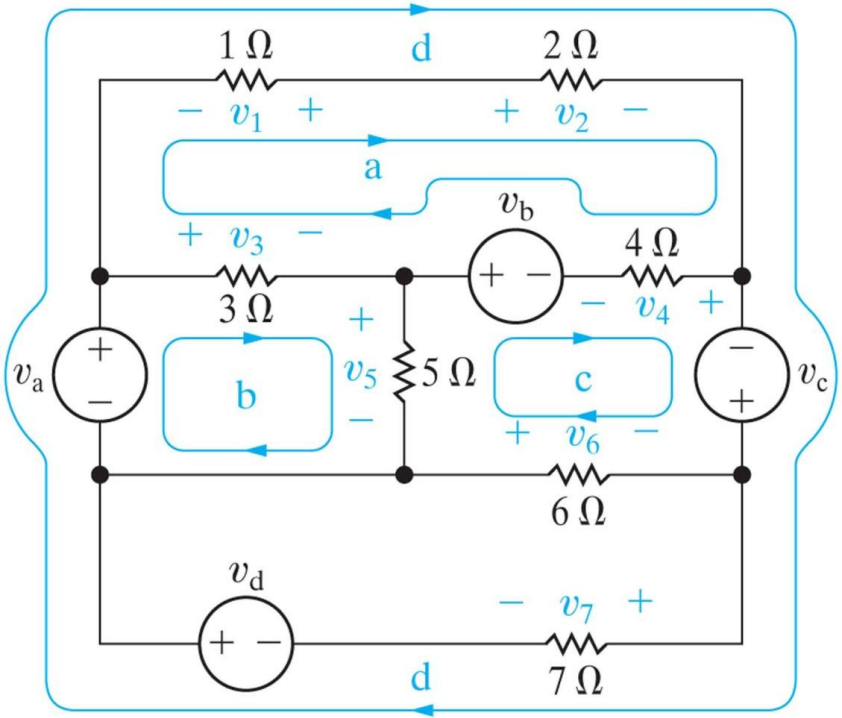


**Continuation to: 4. Kirchoff's Laws**

**5. Analysis of a Circuit Containing Dependent Sources**

# Kirchoff's Laws

- **Example 2**




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KVL following clockwise direction and assigning positive sign to voltage drops:

path a $-v_1 + v_2 + v_4 - v_b - v_3 = 0$

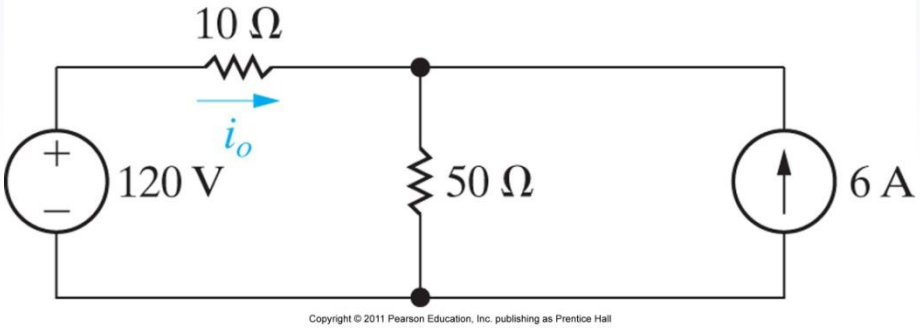
path b $-v_a + v_3 + v_5 = 0$

path c $-v_5 + v_b - v_4 - v_c - v_6 = 0$

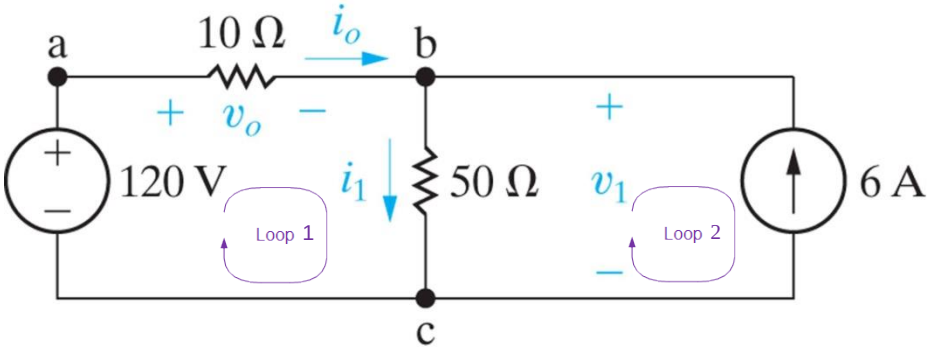
path d $-v_d - v_a - v_1 + v_2 - v_c + v_7 = 0$

# Kirchoff's Laws

- **Example 3**




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Using Kirchoffs laws and Ohm's law, find $i_o$ in the circuit.

We have two unknowns, $i_o$ and $i_1$.

Applying KCL to node b:

node b $\quad -i_o + i_1 - 6 = 0$

Applying Ohm's law:

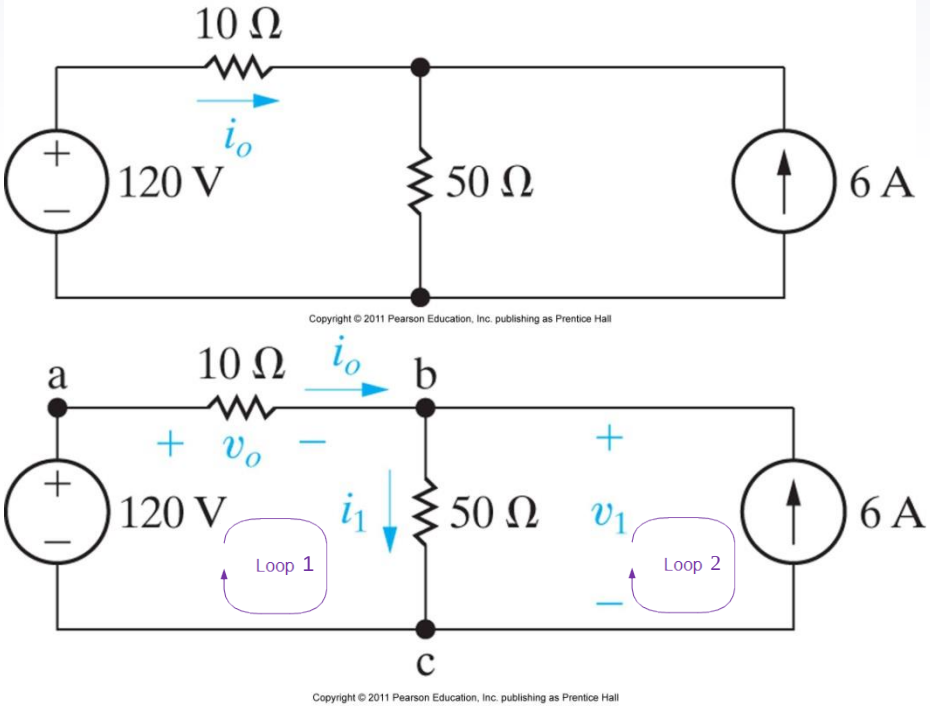
$$v_o = 10i_o \quad \text{and} \quad v_1 = 50i_1$$

Applying KVL on path 1:

$$-120 + 10i_o + 50i_1 = 0$$

# Kirchoff's Laws

- **Example 3**




Solving the two equations with two unknows:

$$(Multiply\ by\ -50) \quad -i_o + i_1 - 6 = 0$$

$$-120 + 10i_o + 50i_1 = 0$$

$$50i_o - 50i_1 + 300 = 0$$

$$10i_o + 50i_1 - 120 = 0$$

$$60i_o = -180$$

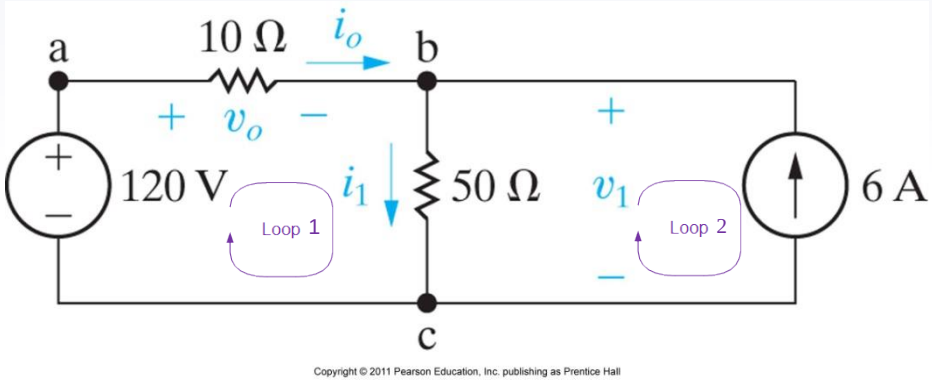
$$i_o = -3\ A$$

$$-(-3) + i_1 - 6 = 0$$

$$i_1 = 3\ A$$

# Kirchoff's Laws

- **Example 3**



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Test the solution for $i_o$ by verifying that the total power generated equals the total power dissipated.

The power dissipated in the 50 Ω resistor

$$p_{50\Omega} = (3)^2\,50 = 450\ W$$

The power dissipated in the 10 Ω resistor

$$p_{10\Omega} = (-3)^2 10 = 90\ W$$

The power delivered to the 120 V source

$$p_{120V} = -(-3)120 = 360\ W$$

The power delivered to the 6A source

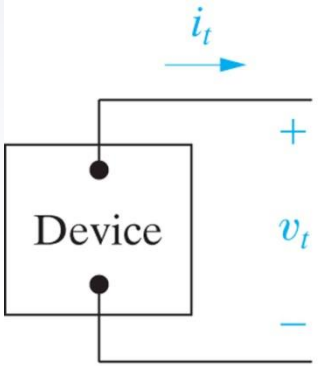
$$p_{6A} = -(6)v_1 \quad v_1 = 50i_1 = 150\ V \quad p_{6A} = -900W$$

The solution verifies that the power delivered is equal to the power absorbed.

# Kirchoff's Laws

4

- **Example 4**




(a)

| $v_t$ (V) | $i_t$ (A) |
|---|---|
| 30 | 0 |
| 15 | 3 |
| 0 | 6 |

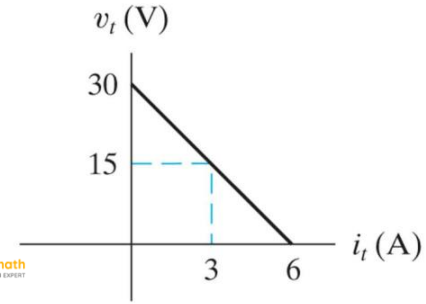
(b)

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The terminal voltage ($v_t$) and terminal current ($i_t$) were measured on the device shown in (a) and their values were given in (b).

Construct a circuit model of the device inside the box.

Plotting the values given in the table (voltage-current characteristic):




$$m = \frac{30 - 15}{0 - 3} = \frac{15}{-3} = -5$$

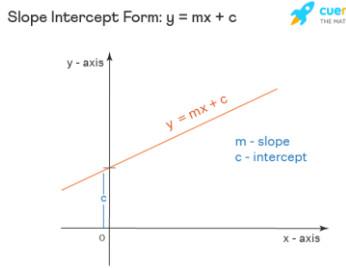
$$c = 30$$

The general equation of any linear line:

$$y = mx + c$$

$m$: gradient

$c$: intercept



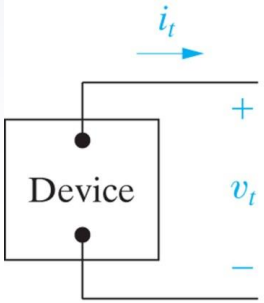

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

The equation of the plotted line:

$$v_t = 30 - 5i_t$$

# Kirchoff's Laws

- **Example 4**

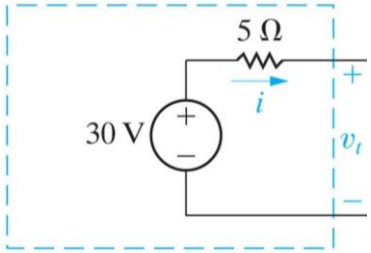


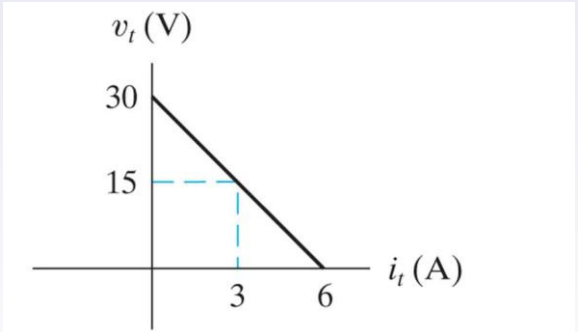

(a)

| $v_t$ (V) | $i_t$ (A) |
|---|---|
| 30 | 0 |
| 15 | 3 |
| 0 | 6 |

(b)

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$$v_t = 30 - 5i_t$$

A component inside the device produces **30 V regardless of the current.**

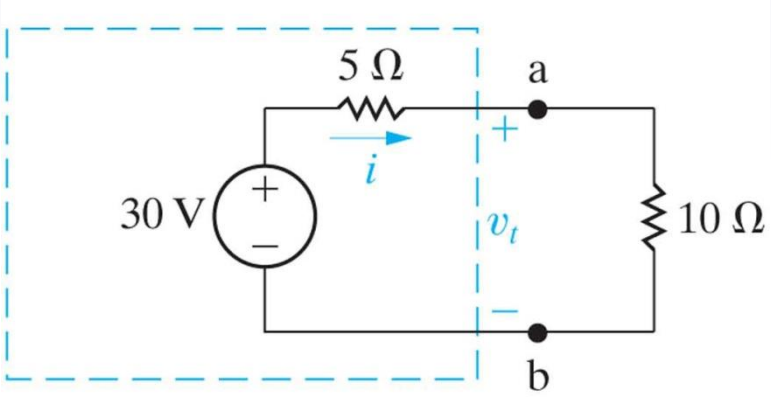
This component can be modelled as an **ideal independent voltage source.**

Another component causes a **positive voltage drop in the direction of the current**.

Because the voltage drop is **proportional** to the current, this component can be modelled according to **Ohm's law** as **an ideal resistor with a value of 5 Ω**.

# 4 Kirchoff's Laws

- **Example 4**




Using the circuit model, calculate the power this device will deliver to a 10 Ω resistor.

Using Kirchoff's Voltage Law, tracing the path clockwise:

$$-30 + 5i + 10i = 0$$

$$15i = 30\ V \qquad i = 2\ A$$

Because this value of current is flowing in the 10 Ω resistor:

$$p = i^2R \qquad p = (2)^2 10 = 40\ \text{W}$$

5

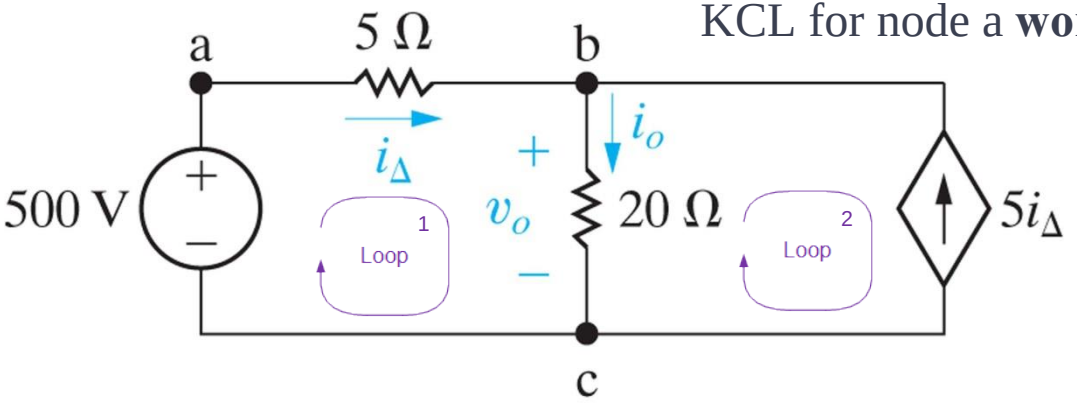
# Circuits Containing Dependent Sources

# Circuits Containing Dependent Sources

- In circuits containing dependent sources the constraints of these sources need also to be taken into account in order to solve the circuit.
- Example: Find $v_o$

We need to find $\boldsymbol{i_o}$ to calculate $v_o$. $\boldsymbol{i_\Delta}$ is needed to find $i_o$.

KCL for node a **won't help**. KVL for path 2 is **not helpful**.




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KCL for node b:

$$-i_\Delta + i_o - 5i_\Delta = 0$$

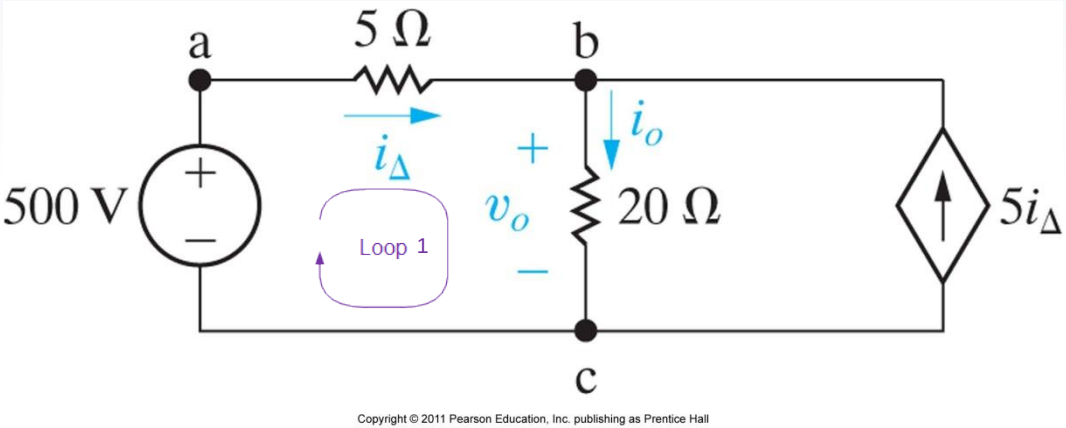
$$i_o = 6i_\Delta$$

KVL for path 1:

$$-500 + 5i_\Delta + 20i_o = 0$$

$$5i_\Delta + 20i_o = 500$$

# 5 Circuits Containing Dependent Sources



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- **Not every node** provides for a useful application of Kirchoff's current law.
- **Not every closed path** provides an opportunity to write a useful equation based on Kirchoff's voltage law.

$$i_o = 6i_\Delta$$

$$5i_\Delta + 20i_o = 500$$

$$5i_\Delta + 20(6i_\Delta) = 500$$

$$125i_\Delta = 500$$

$$i_\Delta = 4\ A$$

$$i_o = 24\ A$$

$$v_o = 20i_o = 480\ V$$

**EE2001 Electrical Circuits- I:**

# Simple Resistive Circuits

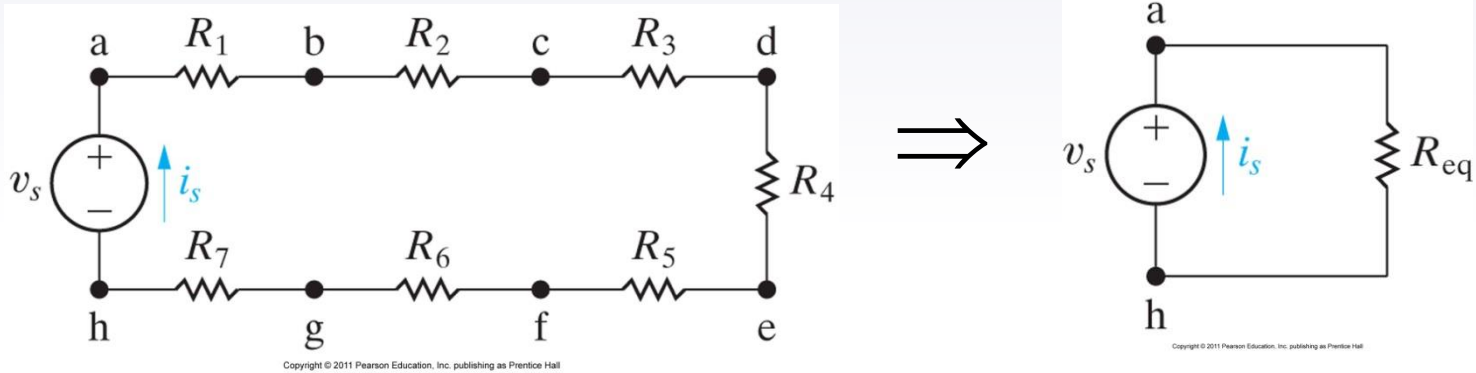
# Outline



1. **Resistors in Series**
2. **Resistors in Parallel**
3. **The Voltage-Divider and Current-Divider Circuits**
4. **Voltage Division and Current Division**

# Resistors in Series

- When two elements **connect at a single node**, they are said to be in **series**.



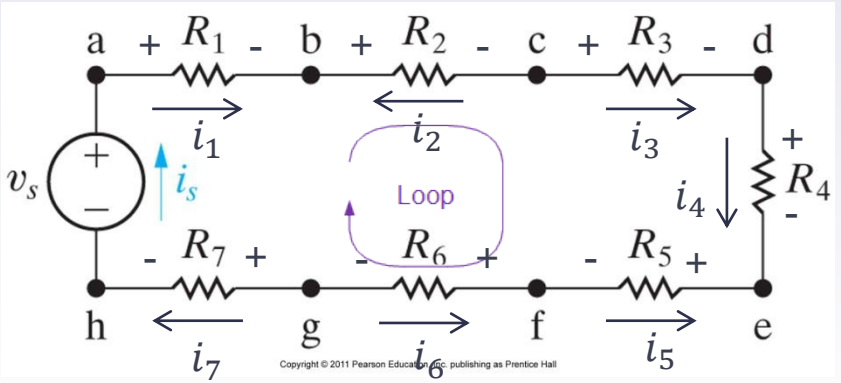

- Series-connected circuit elements **carry the same current** (can be validated with KCL).
- In series connection, the resistance of the **equivalent resistor** is **always larger** than the resistances of the **individual resistors**.

$$R_{eq} = \frac{v_s}{i_s} = R_1 + R_2 + R_3 + \ldots + R_n = \sum_{i=1}^{n} R_i$$

# Resistors in Series

1

- **Derivation** of the equivalent resistor formula for series connected resistances:
  Applying **KCL** (using positive reference direction for each current leaving a node):




| | | | |
|---|---|---|---|
| node a | $-i_s + i_1 = 0$ | $\Rightarrow$ | $i_s = i_1$ |
| node b | $-i_1 - i_2 = 0$ | $\Rightarrow$ | $i_1 = -i_2$ |
| node c | $i_2 + i_3 = 0$ | $\Rightarrow$ | $-i_2 = i_3$ |
| node d | $-i_3 + i_4 = 0$ | $\Rightarrow$ | $i_3 = i_4$ |
| node e | $-i_4 - i_5 = 0$ | $\Rightarrow$ | $i_4 = -i_5$ |
| node f | $i_5 - i_6 = 0$ | $\Rightarrow$ | $i_5 = i_6$ |
| node g | $i_6 + i_7 = 0$ | $\Rightarrow$ | $i_6 = -i_7$ |
| node h | $-i_7 + i_s = 0$ | $\Rightarrow$ | $i_7 = i_s$ |

$$i_s = i_1 = -i_2 = i_3 = i_4 = -i_5 = -i_6 = i_7$$

Applying **KVL** (clockwise):

$$-v_s + i_s R_1 + i_s R_2 + i_s R_3 + i_s R_4 + i_s R_5 + i_s R_6 + i_s R_7 = 0$$

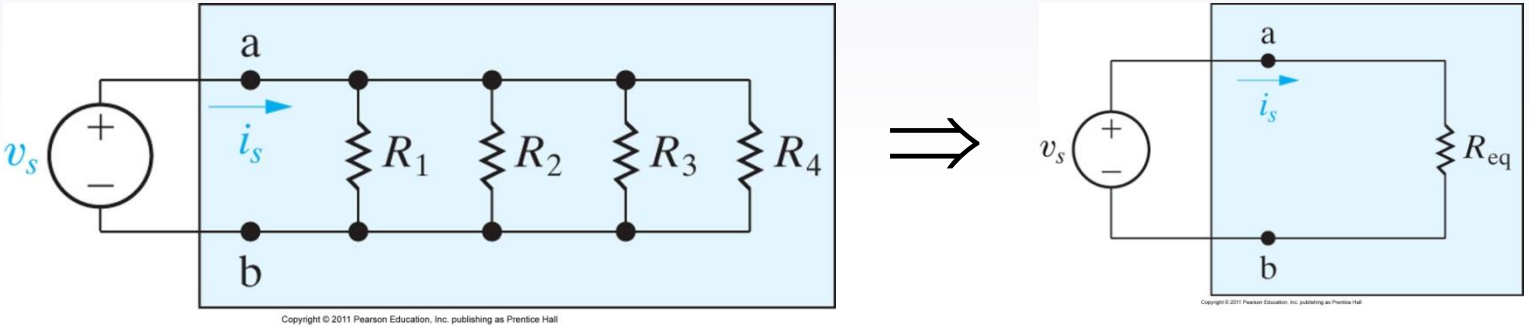
$$v_s = i_s(R_1 + R_2 + R_3 + R_4 + R_5 + R_6 + R_7)$$

For $n$ resistors connected in series:

$$R_{eq} = \frac{v_s}{i_s} = R_1 + R_2 + R_3 + \ldots + R_n = \sum_{i=1}^{n} R_i$$

# 2 Resistors in Parallel

- When two elements **connect at a single node pair**, they are said to be in **parallel**.



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- Parallel-connected circuit elements **have the same voltage** across their terminals.
- In parallel connection, the resistance of the **equivalent resistor** is **always smaller** than the resistances of the **individual resistors**.

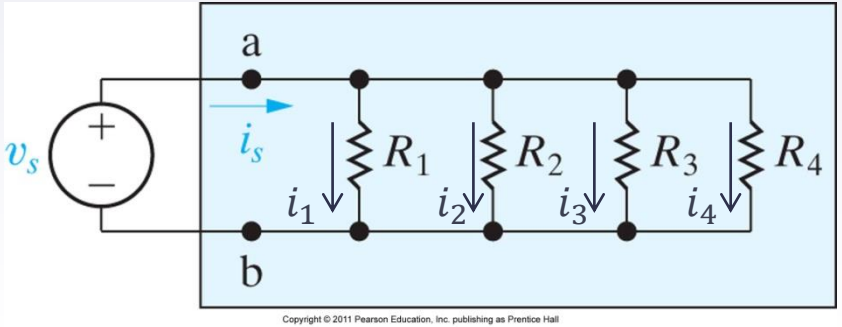
$$\frac{1}{R_{eq}} = \frac{i_s}{v_s} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \ldots + \frac{1}{R_n} = \sum_{i=1}^{n} \frac{1}{R_i}$$

or

$$G_{eq} = \sum_{i=1}^{n} G_i = G_1 + G_2 + \cdots + G_n$$

# 2 Resistors in Parallel

- **Derivation** of the equivalent resistor formula for parallel connected resistances:



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From **KCL** (using positive reference direction for each current leaving a node):

$$i_s = i_1 + i_2 + i_3 + i_4$$

From Ohm's law:

$$i_1 R_1 = i_2 R_2 = i_3 R_3 = i_4 R_4 = v_s \Rightarrow i_1 = \frac{v_s}{R_1} \quad i_2 = \frac{v_s}{R_2} \quad i_3 = \frac{v_s}{R_3} \quad i_4 = \frac{v_s}{R_4}$$

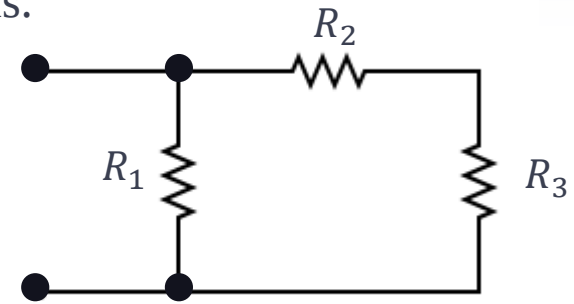
$$i_s = v_s \left( \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \right)$$

For $n$ resistors connected in parallel:

$$\frac{1}{R_{eq}} = \frac{i_s}{v_s} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \ldots + \frac{1}{R_n} = \sum_{i=1}^{n} \frac{1}{R_i}$$

# 2 Resistors in Parallel

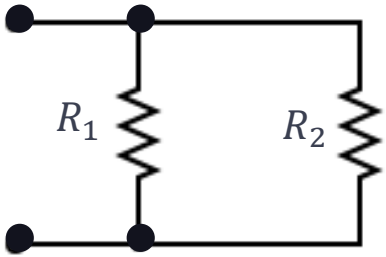
- **Don’t make the mistake of** assuming that two elements are parallel connected **just because** they are **lined up parallel in a circuit diagram**.
- The **defining rule** of parallel connection: they should have **the same voltage** across their terminals.




$R_1$ and $R_3$ are **not parallel** connected, because across their terminals, $\boldsymbol{R_2}$ **dissipates some of the voltage**

# 2 Resistors in Parallel

- **In most cases**, only **two resistors** are connected in parallel.




$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} \rightarrow \frac{1}{R_{eq}} = \frac{R_2}{R_1 R_2} + \frac{R_1}{R_2 R_1} \rightarrow \frac{1}{R_{eq}} = \frac{R_1 + R_2}{R_1 R_2}$$

$$\downarrow$$

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

# End of the Lesson (16.10.23)

## Any questions?

**Supplementary videos**: Enter Youtube-> Search for ‘Alparslan Zehir’ -> Go to Playlists -> Watch ‘Electrical Circuits’ playlist

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